

Question number	Answer	Additional guidance	Mark
5(a)(i)	methods used to prevent contamination (of person / culture) (with other microorganisms) (1)	ACCEPT techniques / procedures / routines ACCEPT prevent {entry / exit / infection} IGNORE growth	(1)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	A description that includes two of the following points: • carry out work beside {a bunsen burner / in a hood} (1) • use sterilised {equipment / media} / sterilise equipment after use (1) • credit named personal procedure (1) • minimise the time that cultures are exposed to the air / do not open cultures at the end (1)	ACCEPT description of how equipment could be sterilised e.g. media autoclaved, wash area with disinfectant e.g. wearing gloves, washing hands e.g. transfer bacteria quickly	(2)

Question number	Answer	Additional guidance	Mark
5(a)(iii)	An explanation that includes two of the following points: • to prevent entry of bacteria that may compete with the <i>E. coli</i> (1) • to prevent entry of bacteria that may grow in different {types / concentrations} of microbial substances (than <i>E. coli</i>) (1) • to prevent infection (of person) with bacteria in the culture that {is / maybe} pathogenic (1)	ACCEPT will not know if <i>E.coli</i> or the other bacteria is growing	(2)

Question number	Answer	Additional guidance	Mark
5(b)	An explanation that includes the following points: • explanation for temperature given (1) • explanation for time given (1)	e.g. appropriate temperature needed for the enzymes to function temperature not too high so the enzymes do not denature ACCEPT proteins for enzymes temp won't be rate limiting for growth e.g. antimicrobial effect can be seen when bacteria are growing enough time has to be allowed for the bacterial growth to become visible enough time for antimicrobials to have an effect	(2)

Question number	Answer	Additional guidance	Mark
5(c)(i)	An explanation that includes the following points: - to show if the cultures were contaminated (1) - because the antimicrobial agent would not {kill / inhibit the growth of} other types of bacteria (1) OR - to check that the <i>E. coli</i> has not changed its {susceptibility / resistance} (1) - so that the results will apply to known <i>E. coli</i> (1)	ACCEPT a description e.g. to see if other bacteria are growing	(2)

Question number	Answer	Additional guidance	Mark
5(c)(ii)	An explanation that includes the following points: - to show that the <i>E.coli</i> were viable (1) - because if there was not any growth of bacteria you would not know if the bacteria were dead or the antimicrobial agents were (very) effective (1)	ACCEPT alive / can replicate / can grow ACCEPT to show that the antimicrobials were inhibiting growth IGNORE antimicrobials affecting growth	(2)

Question number	Answer	Additional guidance	Mark
5(c)(iii)	A description that includes the following points: - description of adding equal volumes of {antimicrobial substance / antimicrobials / substance / solution} and {water / media / broth / buffer} together (1) - description of this being repeated (a few times) using previous solution (1)	ACCEPT the idea of doing this process in the wells directly or separately ACCEPT stated volumes / 50% volumes / volumes in ratio 1 : 1 IGNORE amount ACCEPT a reference to serial dilutions	(2)

Question number	Answer	Additional guidance	Mark
5(c)(iv)	- MIC of E and G calculated (1) {32 / 2⁵} (1)	MIC of E = 1 in 8 and MIC of G = 1 in 256 Or MIC of E = 1 in 16 and MIC of G = 1 in 512 ACCEPT E is {32 / 2⁵} times less effective NB correct answer gains 2 marks ACCEPT 2.25 for 1 mark	(2)

